ULK Marks Appeal System

Chapter Four — System Analysis, Design and Implementation Diagrams

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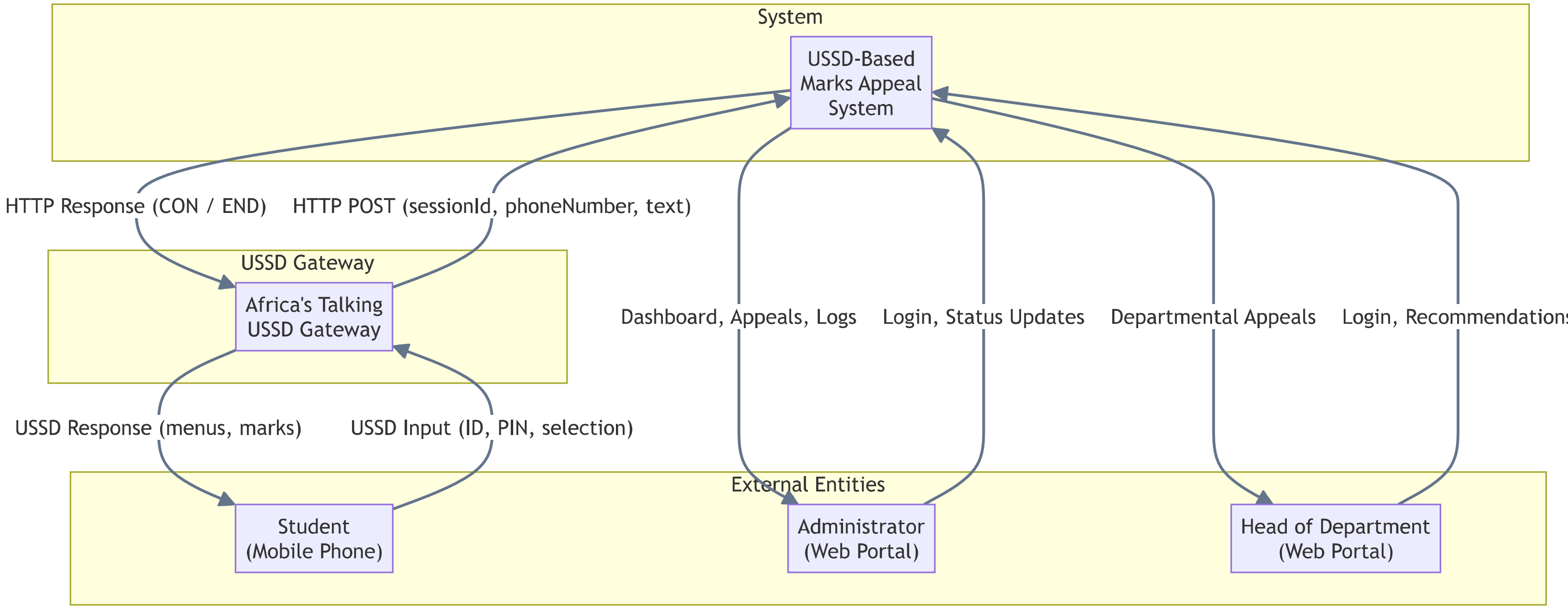
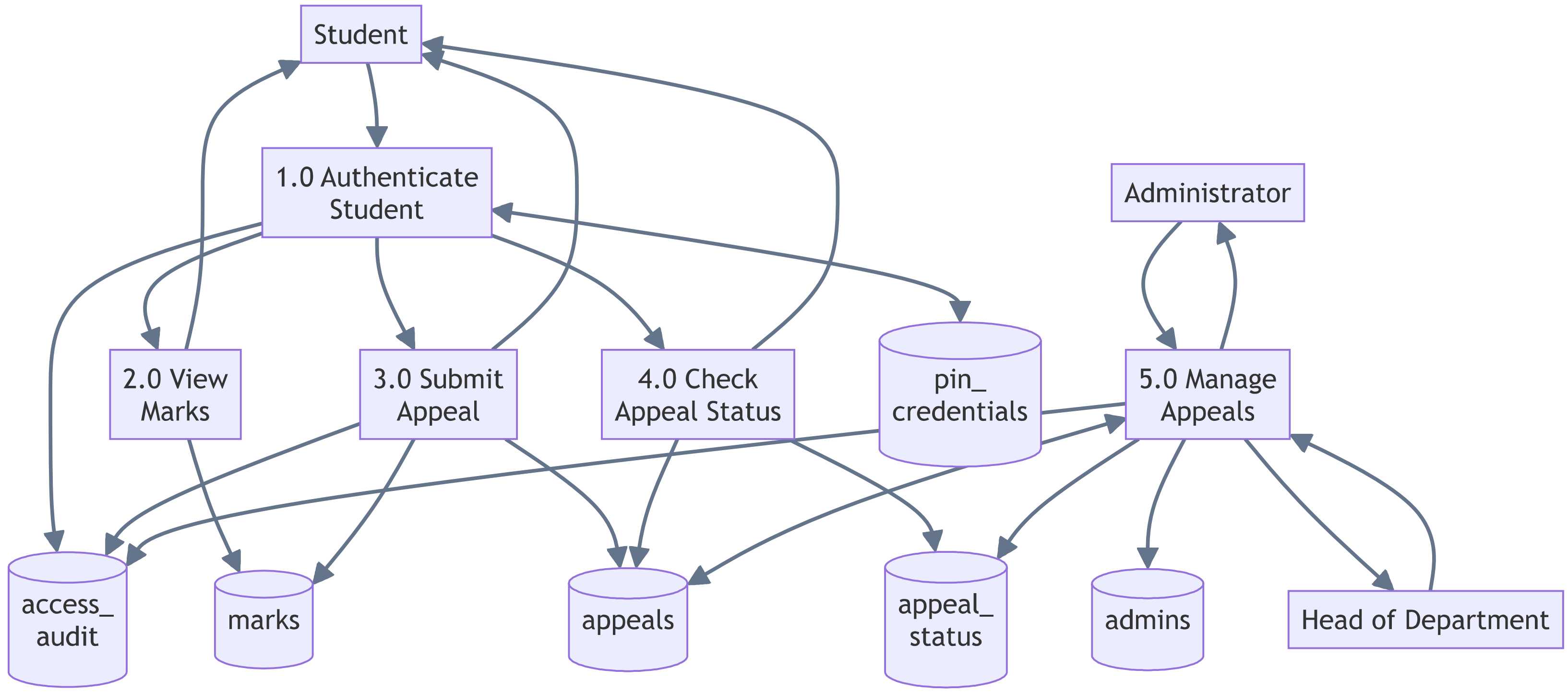
1. Context Diagram (DFD Level 0)

The system as a single process with three external entities: Student (via Africa's Talking USSD Gateway), Administrator, and Head of Department.



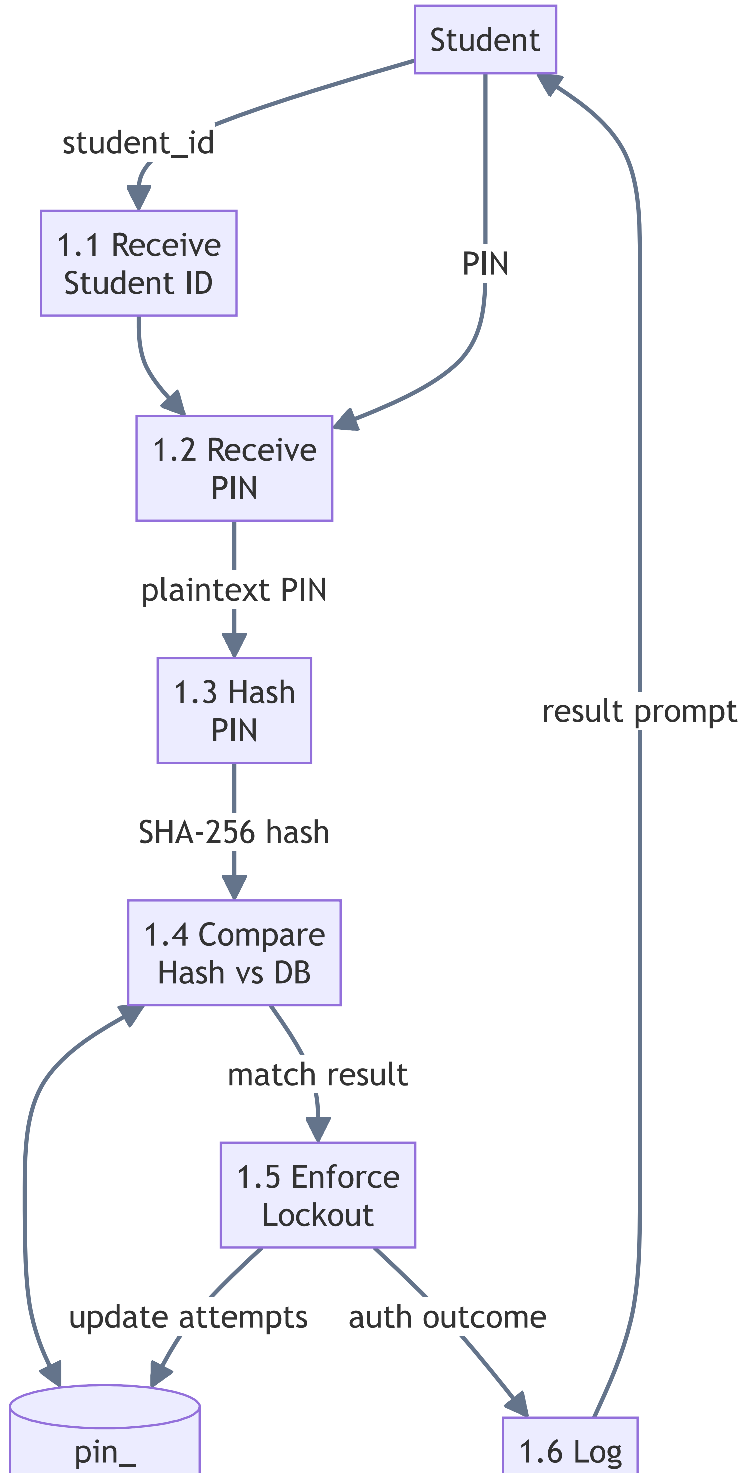
2. Data Flow Diagram — Level 1

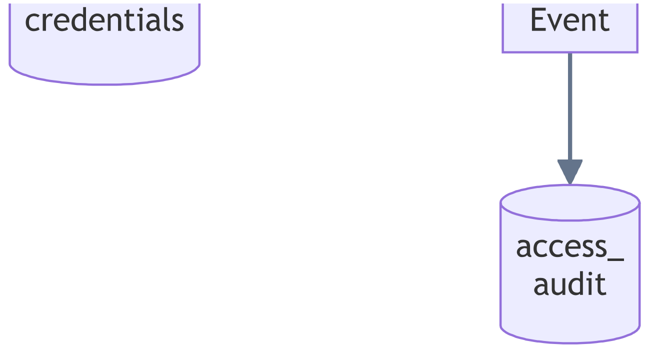
Five primary processes with their data stores: Authenticate Student, View Marks, Submit Appeal, Check Appeal Status, Manage Appeals.



3. Data Flow Diagram — Level 2 (Authenticate Student)

Decomposition of Process 1.0 into six sub-processes: Receive Student ID, Receive PIN, Hash PIN, Compare Hash vs DB, Enforce Lockout Policy, Log Event.


Student
student_id
PIN
1.1 Receive Student ID
1.2 Receive PIN
plaintext PIN
1.3 Hash PIN
SHA-256 hash
1.4 Compare Hash vs DB
match result
1.5 Enforce Lockout
update attempts
auth outcome
result prompt
pin_
1.6 Log



4. Use Case Diagram

Three actors (Student, Administrator, HOD) with 14 use cases. UC4, UC5, UC6 include UC2 (Authenticate via PIN). UC3 extends UC2.

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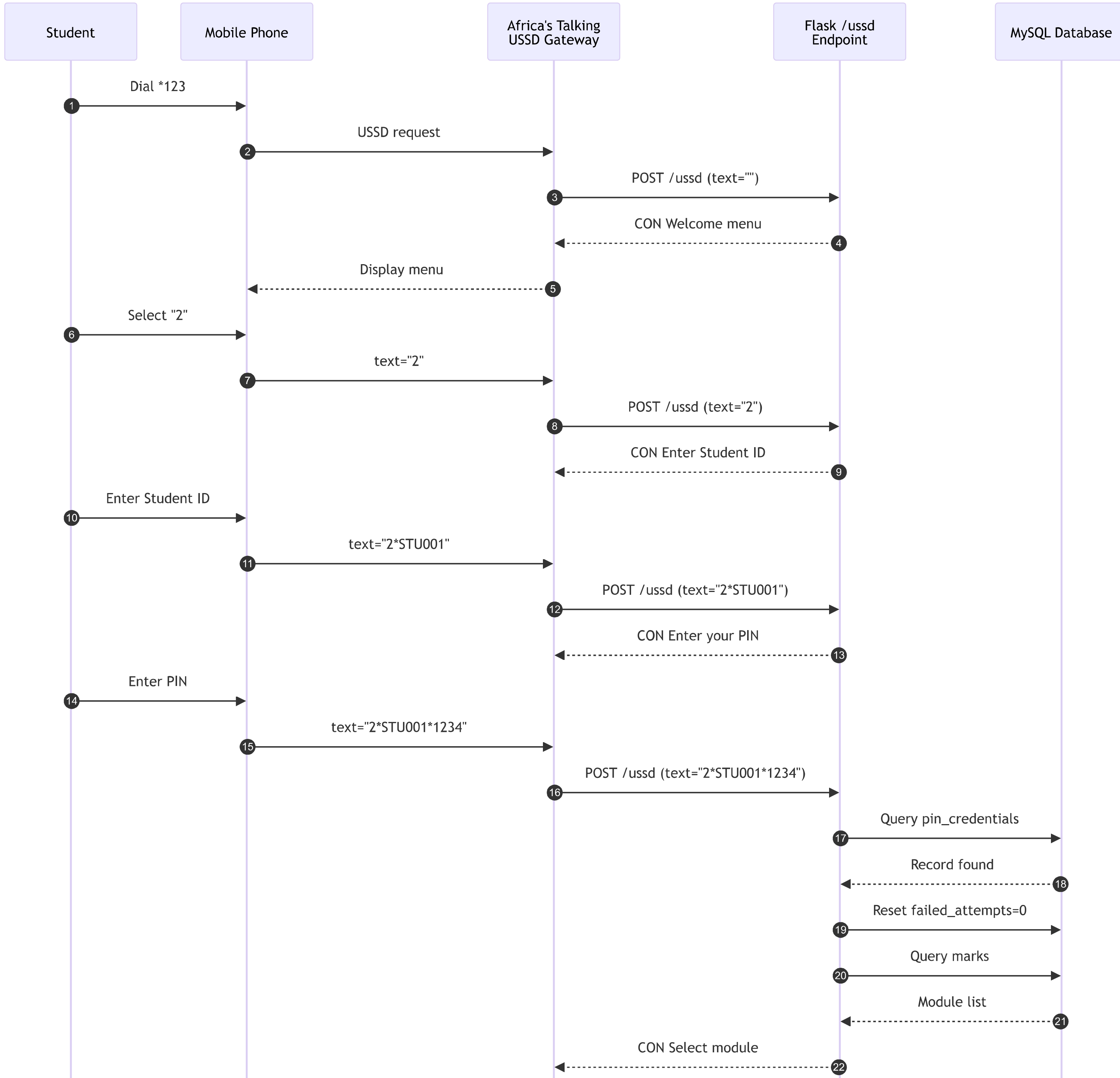
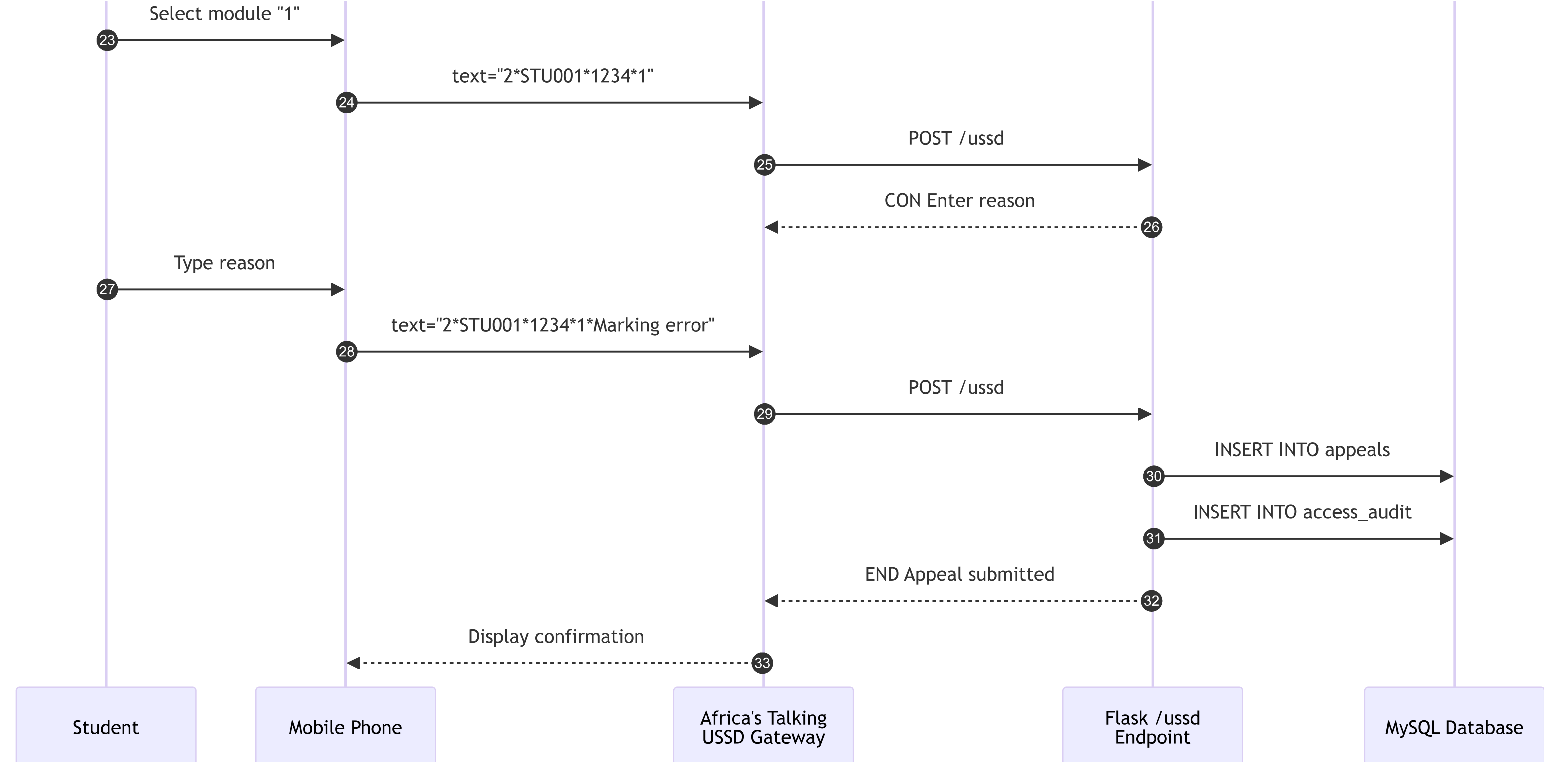
5. Sequence Diagram — Student Submits an Appeal

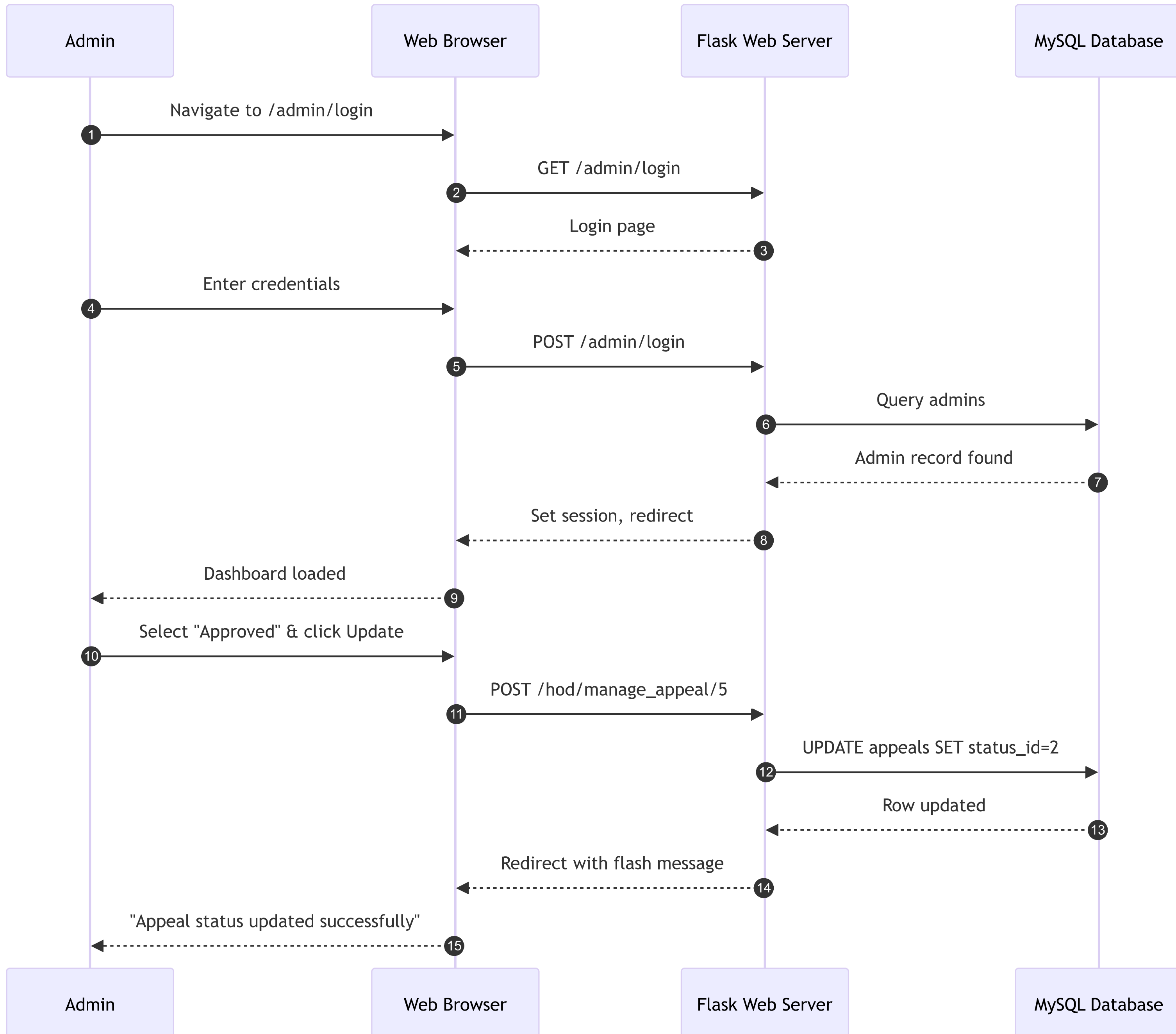
Complete interaction flow showing Student, Africa's Talking Gateway, Flask backend, and MySQL database for the Submit Appeal use case.


Student
Mobile Phone
Africa's Talking USSD Gateway
Flask /ussd Endpoint
MySQL Database
Dial *123
USSD request
POST /ussd (text="")
CON Welcome menu
Display menu
Select "2"
text="2"
POST /ussd (text="2")
CON Enter Student ID
Enter Student ID
text="2*STU001"
POST /ussd (text="2*STU001")
CON Enter your PIN
Enter PIN
text="2*STU001*1234"
POST /ussd (text="2*STU001*1234")
Query pin_credentials
Record found
Reset failed_attempts=0
Query marks
Module list
CON Select module



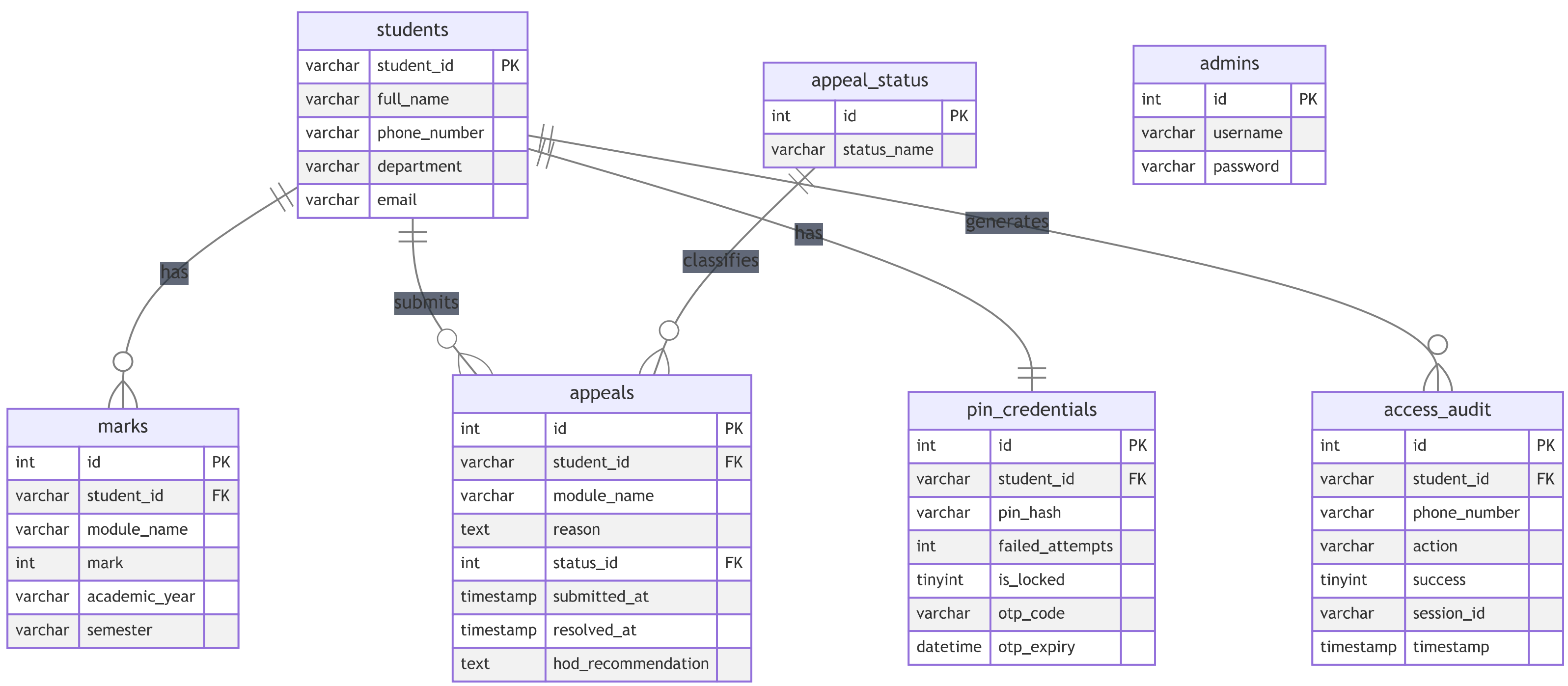
6. Sequence Diagram — Administrator Updates Appeal Status

Interaction showing admin login, dashboard view, and status update flow through Flask to MySQL.


Admin
Web Browser
Flask Web Server
MySQL Database
1 Navigate to /admin/login
2 GET /admin/login
3 Login page
4 Enter credentials
5 POST /admin/login
6 Query admins
7 Admin record found
8 Set session, redirect
9 Dashboard loaded
10 Select "Approved" & click Update
11 POST /hod/manage_appeal/5
12 UPDATE appeals SET status_id=2
13 Row updated
14 Redirect with flash message
15 "Appeal status updated successfully"
Admin
Web Browser
Flask Web Server
MySQL Database

7. Entity Relationship Diagram (ERD)

Seven entities with primary keys, foreign keys, and relationships: students (1) → marks (M), students (1) → appeals (M), students (1) → pin_credentials (1), students (1) → access_audit (M), appeal_status (1) → appeals (M).



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